

Roll No. :S003	Name: Aryan Raj
Class : SYIT	Batch :
Date of Assignment : 19-08-26	Date/Time of Submission : 19-08-26

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Aryan	BSc IT	99	98
1	102	Jasad	BSc CS	65	80
2	103	Kaif	BSc IT	88	92
3	104	Sopte	BSc CS	72	80
4	105	Dhruv	BSc IT	55	70

```
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Aryan	BSc IT	99	98
1	102	Jasad	BSc CS	65	80
2	103	Kaif	BSc IT	88	92
3	104	Sopte	BSc CS	72	80
4	105	Dhruv	BSc IT	55	70

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.shape) #number of rows and columns
```

```
(5, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

```
result = df[df["Attendance"]>=90]
print(result)
```

	Roll No	Name	Department	Marks	Attendance
0	101	Aryan	BSc IT	99	98
2	103	Kaif	BSc IT	88	92

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

```
result = df.sort_values("Marks")
print(result)
```

	Roll No	Name	Department	Marks	Attendance
4	105	Dhruv	BSc IT	55	70
1	102	Jasad	BSc CS	65	80
3	104	Sopte	BSc CS	72	80
2	103	Kaif	BSc IT	88	92
0	101	Aryan	BSc IT	99	98

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks: ",df["Marks"].max())
print("Minimum Marks: ",df["Marks"].min())
print("Average Marks: ",df["Marks"].mean())
print("Total Marks: ",df["Marks"].sum())
```

```
Maximum Marks: 99
Minimum Marks: 55
Average Marks: 75.8
Total Marks: 379
```